WorkPulse

System Specification Summary

Project overview / architecture / workflows / security

1. Project overview

WorkPulse is a proposed web-based employee attendance and workforce analytics information system for MVL. It brings together attendance records, employee information, leave requests, payroll-related summaries and bounded machine-learning insights in a role-based interface.

Users and responsibilities

User	Main capabilities
Employee	Sign in, view personal attendance and payroll-related information, and submit leave requests where permitted.
Administrator / Owner	Maintain employee records, review attendance and corrections, process leave requests, inspect payroll information and reports, and access authorized analytics.

Essential system modules

Module	Important functions
Accounts and employee directory	Role-based login, employee records, account status and administrative controls.
Attendance and biometrics	Fingerprint-assisted time-in / time-out, attendance history, late or missed-scan review and corrections.
Leave management	Employee requests, administrator review, decision status and leave history.
Payroll and reporting	Payroll-related records and summaries; calculations and policy-dependent components require MVL confirmation.
AI Insights	Seven-day attendance-volume forecasting, 30-day attendance-risk estimates and unusual-arrival detection.
Settings and oversight	Schedule / policy configuration, system settings and controlled administrative actions.

System boundary

The web interface communicates with the application backend and MongoDB. A fingerprint device is used for biometric attendance functions. A separate private Python service supports the bounded machine-learning modules; statistical fallback is identified separately when a trained model is unavailable.

This document summarizes known project scope and prior design decisions. Unverified deployment details, completed testing results and MVL policies are not asserted as finalized.

2. System architecture

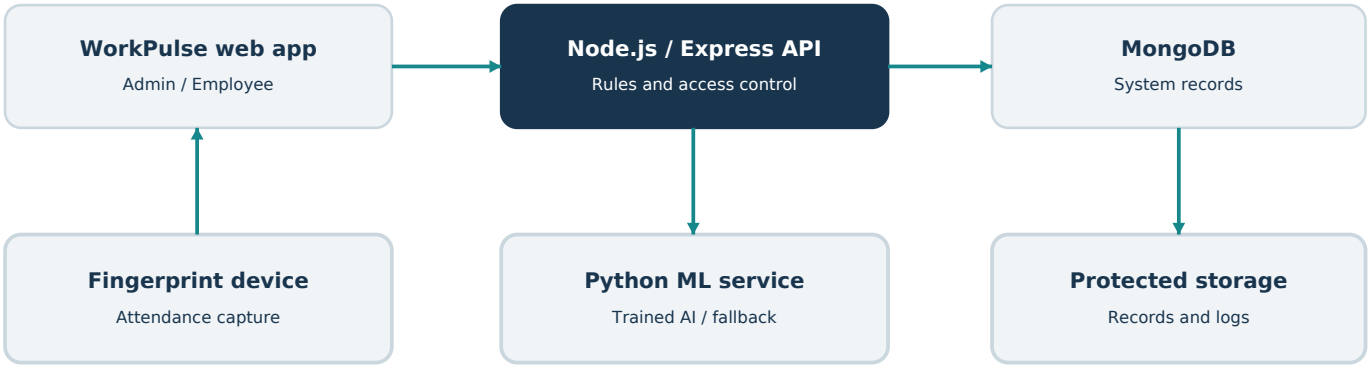


Figure 1. Conceptual WorkPulse components and their principal connections. Exact deployment topology and device integration must be verified against the implemented system.

Technology stack

Component	Technology / approach	Purpose
Web interface	Browser-based frontend; exact framework to verify	Employee and administrator dashboards.
Backend API	Node.js; watch mode during development	Authentication, validation, system rules and data operations.
Database	MongoDB / MongoDB Atlas	Employee, attendance, leave and payroll-related records.
Password protection	Node.js native scrypt with random salts	Protect stored password verifiers.
Biometric attendance	Fingerprint scanner and matching integration	Attendance capture and verification; not an AI Insights model.
Machine learning	Private Python service; three bounded models	Forecasting, risk scoring and unusual-arrival detection.
Development / hosting	Project configuration; AWS only if deployed	Runtime and access to WorkPulse services.

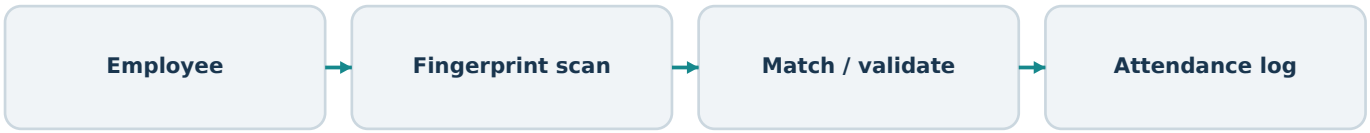
AI implementation boundary

A result is identified as Trained AI only when the interface provides a model version, training date and sample count. If the model cannot run or sufficient data is unavailable, the interface must identify a Statistical fallback. WorkPulse does not use a generative chatbot or large language model for these insights.

Fingerprint recognition and scanner status belong to biometric evaluation, not to the three AI Insights modules.

3. Core workflows and system rules

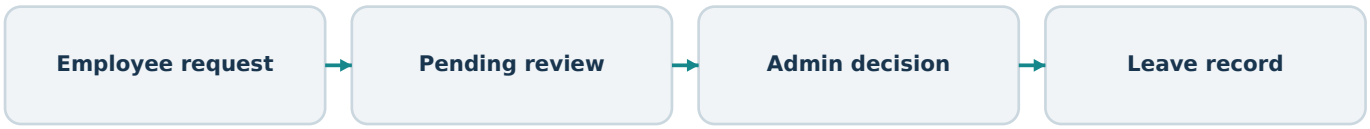
Attendance recording



Failed or missing scans require the designated correction / review process.

Figure 2. Basic attendance flow. The system validates the biometric event before storing a time-in or time-out record.

Leave request handling



Approval or rejection is recorded; pay treatment follows confirmed MVL policy.

Figure 3. Leave requests pass through a review decision before the resulting status is recorded.

Key operating rules

Rule	Required / proposed behavior
Authentication	Login validates required fields, CAPTCHA, credentials, session state, account status and role-based access.
Attendance ownership	Proposed policy: employees record their own attendance; recording attendance for another person is prohibited.
Lateness	Determined from the assigned schedule and any grace period approved by MVL.
Missed scans / corrections	Exceptions are reported and reviewed by the designated authorized person.
Leave	Eligibility, allowance and paid / unpaid treatment depend on MVL-approved policy.
Payroll	Overtime, holiday pay, night differential, deductions and 13th-month pay require verified rules and designated review.
AI results	Predictions are decision-support information, not automatic disciplinary decisions or guaranteed outcomes.

Policy note: these attendance, leave and payroll items include proposed rules that require owner confirmation; they do not automatically establish MVL policy.

4. Data, security and operating essentials

Core data relationships

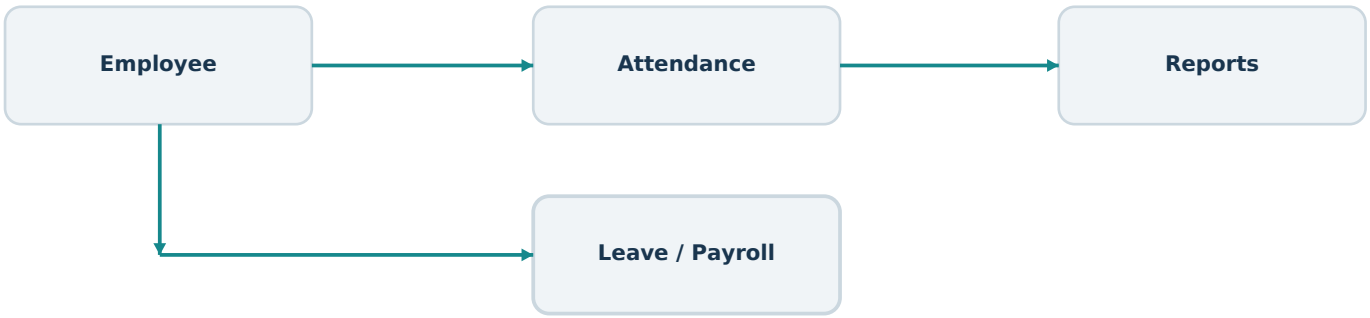


Figure 4. Conceptual relationship between employees, attendance records, leave / payroll records and reports. Detailed cardinalities depend on the database schema.

The database may hold employee profiles, account status, assigned schedules, attendance events, leave requests, payroll-related information and records needed for authorized reporting. Biometric templates, if stored, require separate safeguards and documented retention rules.

Security and privacy

Area	Main controls / requirements
Accounts and access	Salted scrypt password verification, CAPTCHA during login, account-status checks and administrator / employee role authorization.
Data handling	Input validation, least-privilege access and appropriate protection of employee and payroll records.
Biometric information	Authorized enrollment and matching, participant consent where applicable, restricted access and defined retention / deletion.
Audit and corrections	Traceable review of attendance corrections, leave decisions and sensitive administrative changes; implementation must be confirmed.
AI and transparency	Clear trained-model versus statistical-fallback labels; predictions should not be misrepresented as verified facts.

Operating requirements

Users require a supported browser, connectivity to the hosted application and appropriate account permissions. Fingerprint attendance requires a compatible device, working driver / SDK and validated matching integration. The backend requires its configured Node.js runtime and MongoDB connection; machine-learning functionality requires the configured private Python service and model assets.

Evaluation and verification limits

Specific uptime guarantees, concurrent-user capacity, fingerprint FAR / FRR, final payroll automation and actual deployment configuration are not established by this summary. Usability and biometric evaluation results should be reported only from completed and documented tests.

Source basis: WorkPulse project scope, prior requirements, login flow, AI Insights boundaries and proposed MVL policy checklist. This summary is not a certification of live deployment or completed implementation.